Financial Analysis and Decision Making
Global BBA Fall 2026

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Session 7

BUSINESS SCHOOL

SKEMA

Housekeeping

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Historical averages and volatility

Suppose we have the following table of past annual returns for a stock.

Year	Returns
2026	13%
2025	−4%
2024	6%
2023	5%

The average return over some fixed horizon is the simple average,

$$\bar{R} = \frac{1}{T}(R_1 + R_2 + \cdots + R_T) = \frac{1}{T}\sum_{t=1}^{T} R_t$$

Historical averages and volatility

Suppose we have the following table of past annual returns for a stock.

Year	Returns
2026	13%
2025	−4%
2024	6%
2023	5%

The volatility, or population standard deviation of realized returns, measures how much a stock's returns have fluctuated in the past.

$$\bar{\sigma} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(R_t - \bar{R})^2}$$

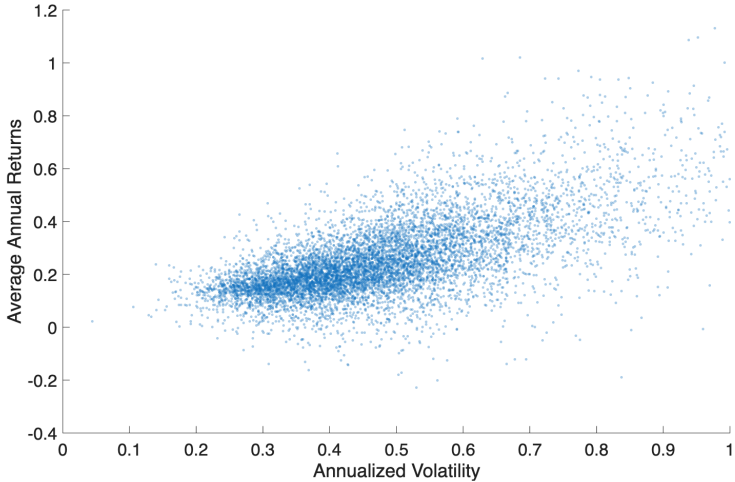
*This formula is identical to the one we saw before, except now we are treating the probability of each realization being equally representative of the distribution.

Plan For Today

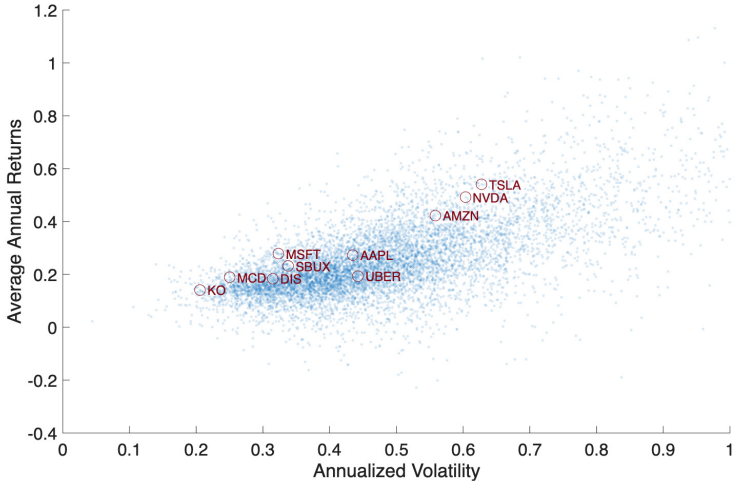


- Some Facts
- Portfolio Mean and Variance
- The Efficient Frontier
- Additional Exercises

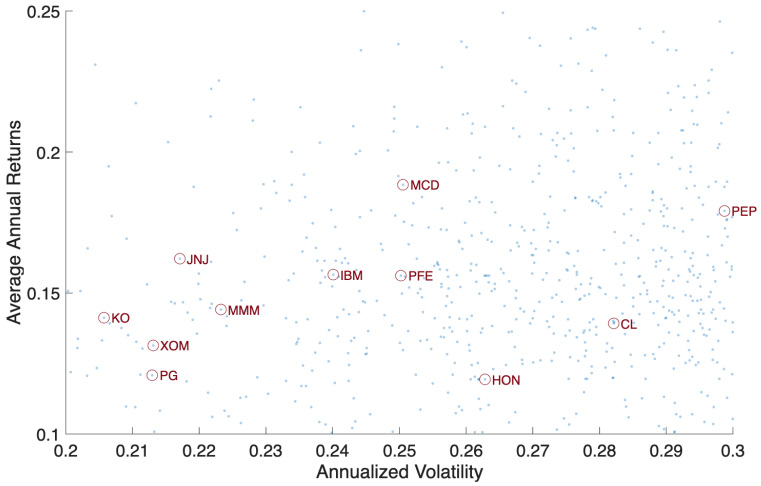
Universe of Stocks: 1926-2025



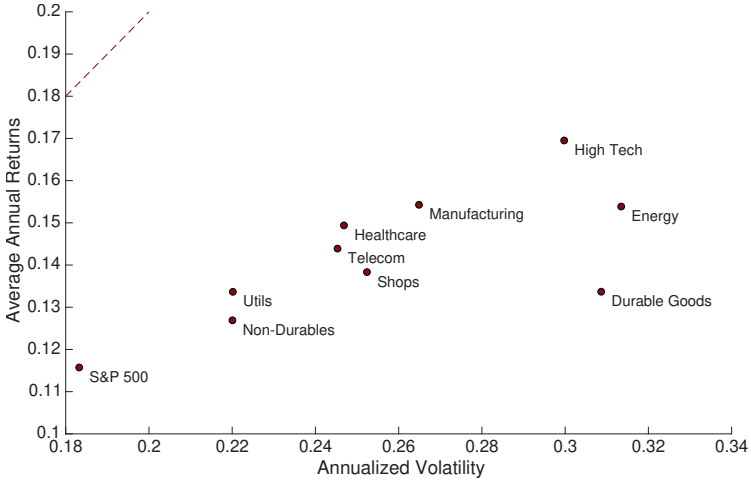
Universe of Stocks: 1926-2025



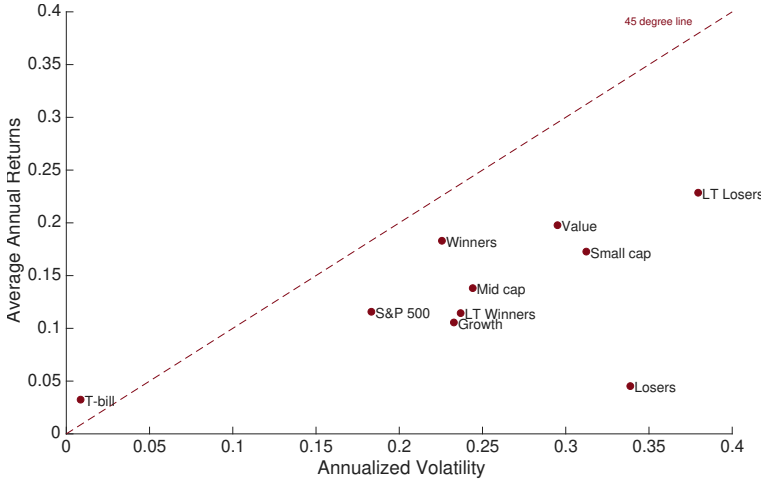
Universe of Stocks: 1926-2025



Industry Portfolios: 1926-2025



Modern Portfolio Choice in 2026



Plan For Today



- Some Facts
- Portfolio Mean and Variance
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- Additional Exercises

Portfolio Returns

Suppose we estimate the following in the data:

Stock	Avg Returns	Avg Vol
KO	9%	15%
AMZN	30%	30%

If we invest 40% of our wealth in KO and the remaining 60% in AMZN, what is the expected return of this portfolio?

Portfolio Returns

Suppose we estimate the following in the data:

Stock	Avg Returns	Avg Vol
KO	9%	15%
AMZN	30%	30%

If we invest 40% of our wealth in KO and the remaining 60% in AMZN, what is the expected return of this portfolio?

$$\begin{aligned}\text{Returns}_{\text{portfolio}} &= w \times \text{Returns}_{\text{KO}} + (1-w) \times \text{Returns}_{\text{AMZN}} \\ &= 0.40 \times .09 + 0.60 \times .30 \\ &= 0.036 + 0.18 = 21.6\%\end{aligned}$$

where w is the weight in KO.

Portfolio Returns

For the moment we will assume that expected returns = average historical returns, though we will revise this assumption in the next class.

The expected returns of a portfolio is given by

$$\mathbb{E}[R_p] = \sum_i w_i \times \mathbb{E}[R_i] \quad \text{such that} \quad \sum_i w_i = 1$$

$\mathbb{E}[R_p]$: expected return of the portfolio

$\mathbb{E}[R_i]$: expected return on asset i

w_i: portfolio weight invested in asset i (as a fraction)

Covariance between Two Assets



Recall, volatility measures how much a stock's returns deviate from its average. When we invest in a portfolio of many assets, we want to know whether one asset does poorly when another asset does poorly, and vice-versa.

Covariance measures how two random variables move together relative to their own averages.

$$\text{cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

and when estimated in sample with T observations,

$$\text{cov}(X, Y) = \frac{1}{T-1} \sum_{t=1}^{T} (X_t - \mathbb{E}[X])(Y_t - \mathbb{E}[Y])$$

Covariance between Two Assets



Suppose we have the following table of past annual returns for KO and AMZN. Assume the expected returns on KO is 9% and the expected returns on AMZN is 30%.

Year	KO	AMZN
2026	25%	10%
2025	15%	5%
2024	9%	40%
2023	−5%	80%

Calculate the sample covariance between KO and AMZN stock returns between 2023-2026.

Sample Covariance

Suppose we have the following table of past annual returns for KO and AMZN. Assume the expected return on KO is 9% and the expected return on AMZN is 30%.

Year	KO	AMZN	KO-$\mathbb{E}$[KO]	AMZN-$\mathbb{E}$[AMZN]	Product
2026	0.25	0.10	0.16	-0.20	-0.0320
2025	0.15	0.05	0.06	-0.25	-0.0150
2024	0.09	0.40	0.00	0.10	0.0000
2023	-0.05	0.80	-0.14	0.50	-0.0700
				Sum of products	-0.1170
				Covariance $= -0.1184/3$	-0.0390

Correlation

The correlation between two random variables normalizes the covariance to a number between -1 and 1. It tells you how strongly the two variables move together,

$$\text{Correlation} = \rho = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

- $\rho = +1$: A perfect positive linear relationship
- $\rho = 0$: No linear relationship
- $\rho = -1$: A perfect negative linear relationship.

Correlation

We found that $\text{cov}(\text{KO}, \text{AMZN}) = -.0390$. Assume $\sigma_{\text{KO}} = 15\%$ and $\sigma_{\text{AMZN}} = 30\%$. What is the correlation between KO and AMZN?

$$\text{Correlation} = \rho = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

Plugging in the numbers,

$$\rho = \frac{-0.0390}{.15 \times .30} = -0.87$$

In the recent data, KO and AMZN tend move in opposite directions.

Portfolio Variance of Two Risky Assets



Let *w* be the weight invested in KO and (1-w) be the weight invested in AMZN. The portfolio variance is given by

$$\text{var}(p) = \sigma_p^2 = w^2\sigma_{\text{KO}}^2 + (1-w)^2\sigma_{\text{AMZN}}^2 + 2w(1-w)\text{cov}(\text{KO}, \text{AMZN})$$

which is equal to

$$\text{var}(p) = \sigma_p^2 = w^2\sigma_{\text{KO}}^2 + (1-w)^2\sigma_{\text{AMZN}}^2 + 2w(1-w)\rho\sigma_{\text{KO}}\sigma_{\text{AMZN}}$$

Portfolio Variance of Two Risky Assets



If we invest 40% of our wealth in KO and the remaining 60% in AMZN, what is the volatility of this portfolio?
Assume $\rho = -0.87$ and $\sigma_{\text{KO}} = 15\%$ and $\sigma_{\text{AMZN}} = 30\%$.

$$\text{var}(p) = \sigma_p^2 = w^2\sigma_{\text{KO}}^2 + (1-w)^2\sigma_{\text{AMZN}}^2 + 2w(1-w)\rho\sigma_{\text{KO}}\sigma_{\text{AMZN}}$$

Portfolio Variance of Two Risky Assets



If we invest 40% of our wealth in KO and the remaining 60% in AMZN, what is the volatility of this portfolio?
Assume $\rho = -0.87$ and $\sigma_{\text{KO}} = 15\%$ and $\sigma_{\text{AMZN}} = 30\%$.

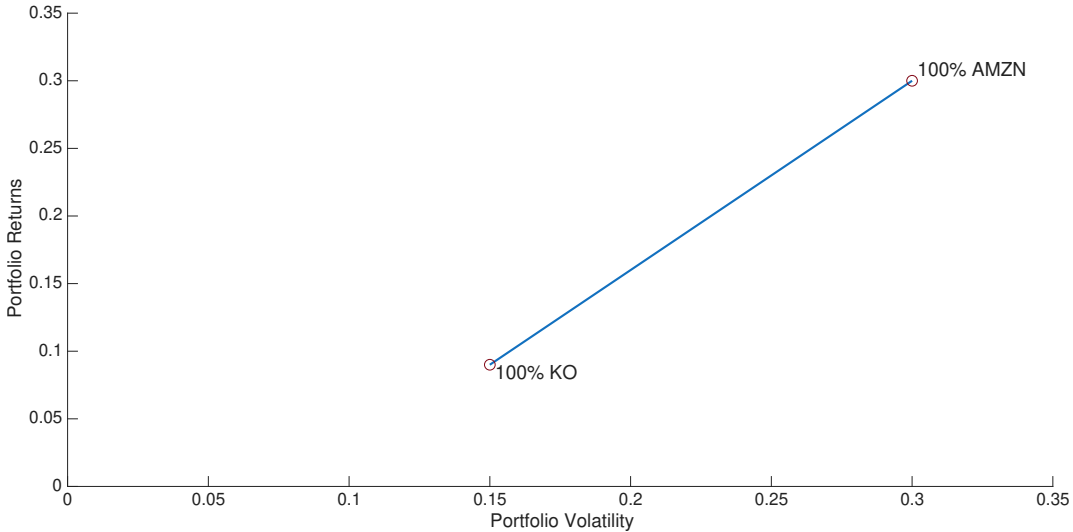
$$\text{var}(p) = \sigma_p^2 = w^2\sigma_{\text{KO}}^2 + (1-w)^2\sigma_{\text{AMZN}}^2 + 2w(1-w)\rho\sigma_{\text{KO}}\sigma_{\text{AMZN}}$$

Plugging in,

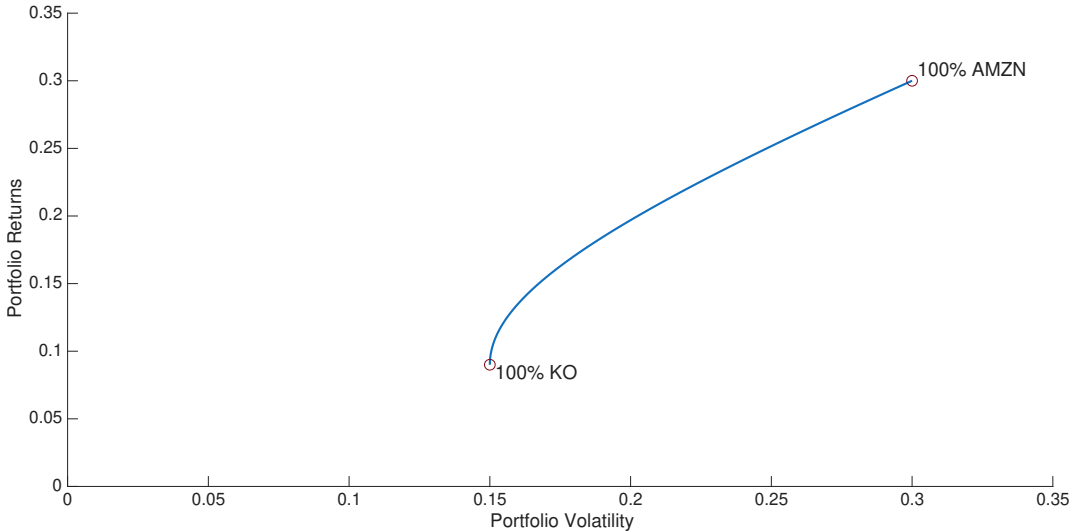
$$\sigma_p^2 = 0.4^2(0.15)^2 + 0.6^2(0.30)^2 + 2(0.4)(0.6)(-0.87)(0.15)(0.30) = 0.0172$$
$$\sigma_p = \sqrt{0.0172} \approx 13.1\%$$

Recall, the expected returns on this portfolio was 21.6%. Due to the strong negative correlation between KO and AMZN, our portfolio volatility is lower than the volatility of KO despite having more than twice the expected returns.

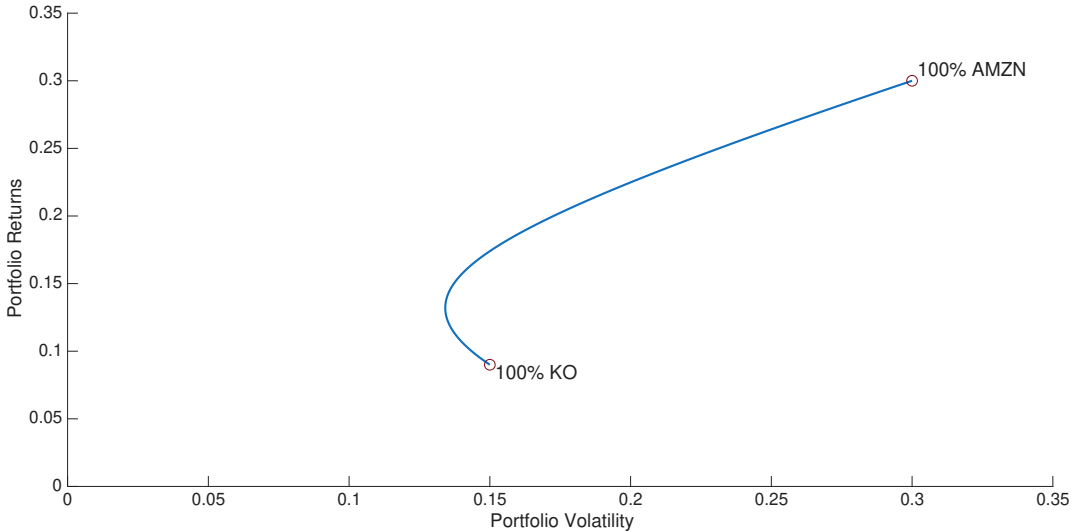
Risk-Return Plane with Two Risky Assets, $\rho = 1$



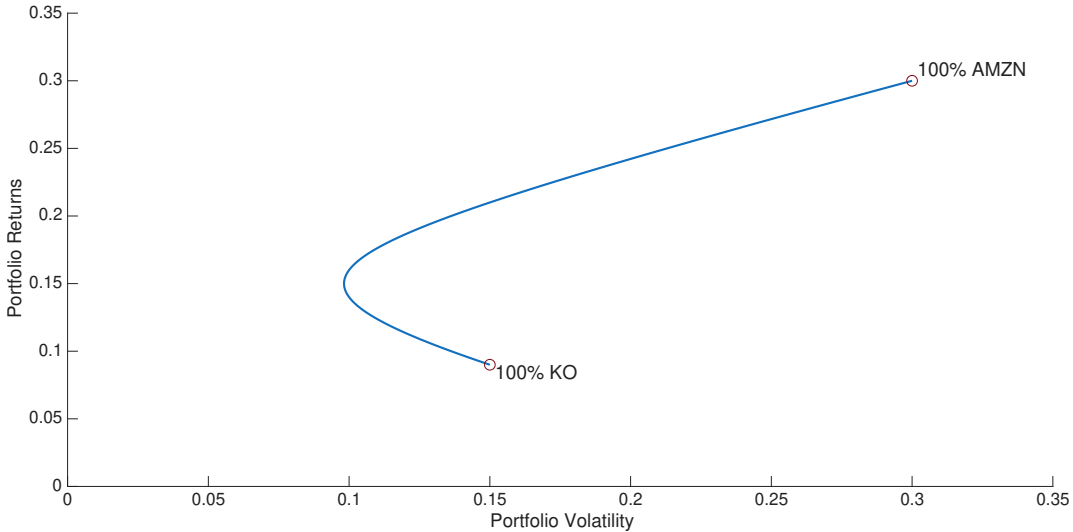
Risk-Return Plane with Two Risky Assets, $\rho = 0.5$



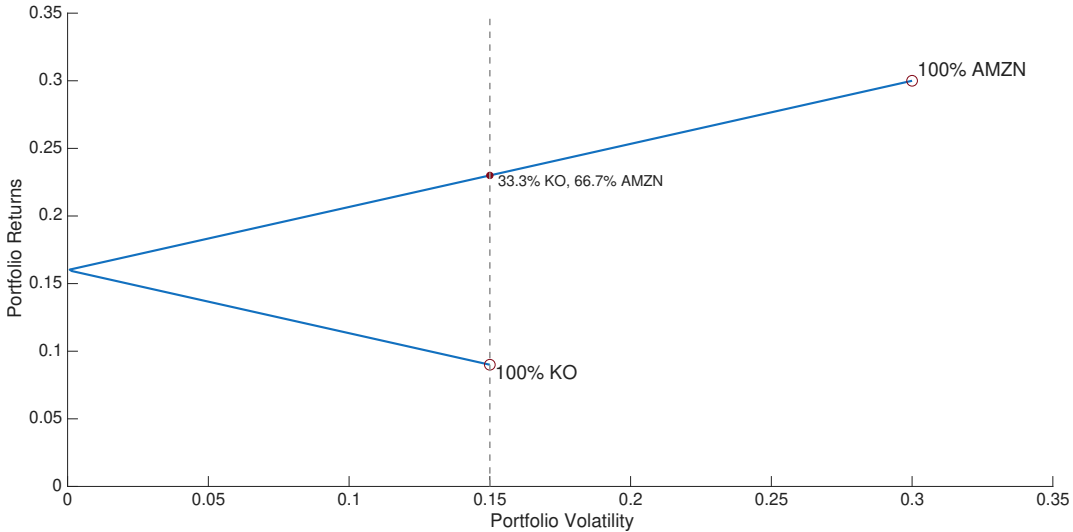
Risk-Return Plane with Two Risky Assets, $\rho = 0$



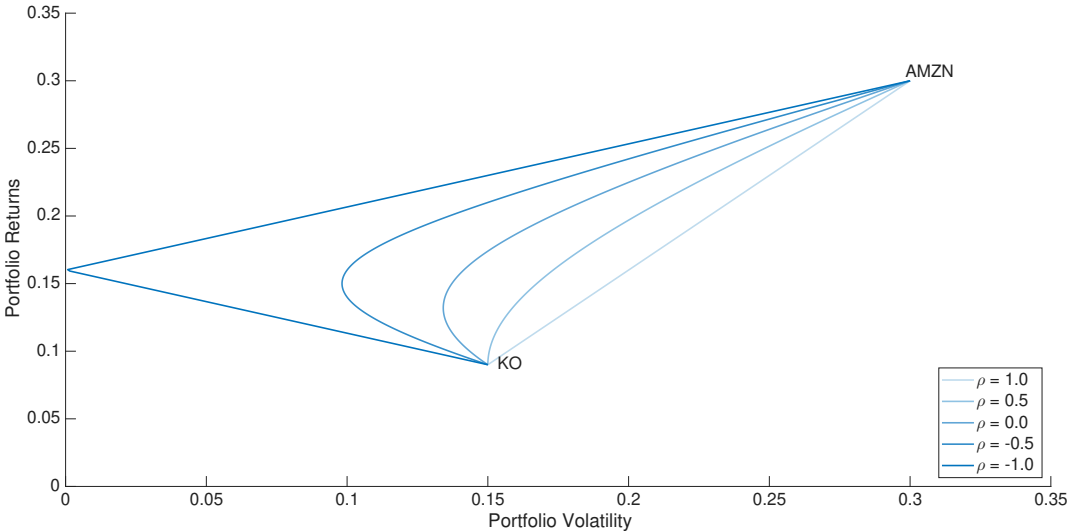
Risk-Return Plane with Two Risky Assets, $\rho = -0.5$



Risk-Return Plane with Two Risky Assets, $\rho = -1$



Risk-Return Plane with Two Risky Assets

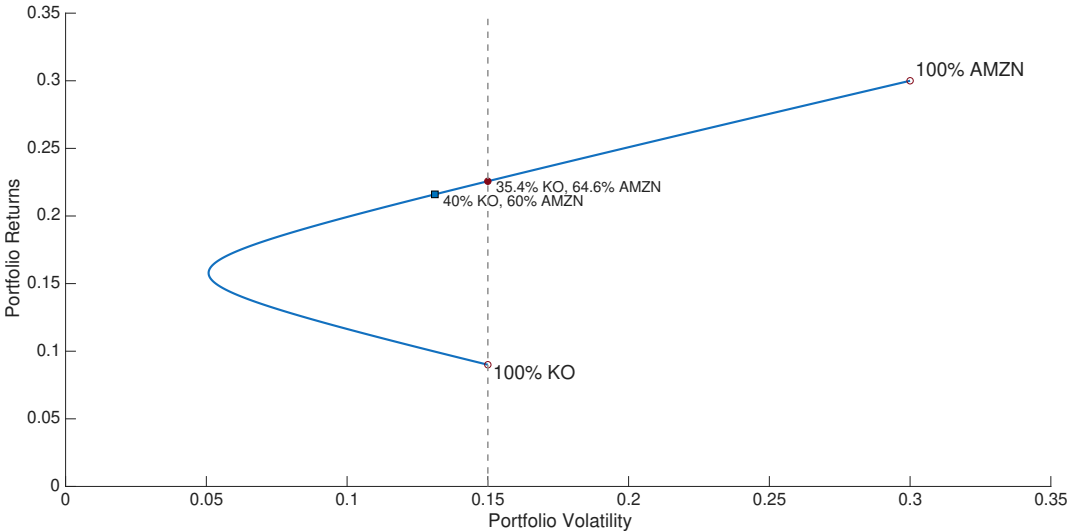


Plan For Today



- Some Facts
- Portfolio Mean and Variance
- The Efficient Frontier
- Additional Exercises

Risk-Return Plane with Two Risky Assets, $\rho = -0.87$



The Efficient Frontier



KO had an average return of 9% and volatility of $\sigma_{KO} = 15\%$. A portfolio of 35.4% in KO and 64.6% in AMZN gives
Portfolio returns

$$\begin{aligned}\mathbb{E}[R_p] &= w\mathbb{E}[R_{KO}] + (1-w)\mathbb{E}[R_{AMZN}] \\ &= .354 \times .09 + .646 \times .3 \\ \mathbb{E}[R_p] &= 22.6\%\end{aligned}$$

Portfolio volatility,

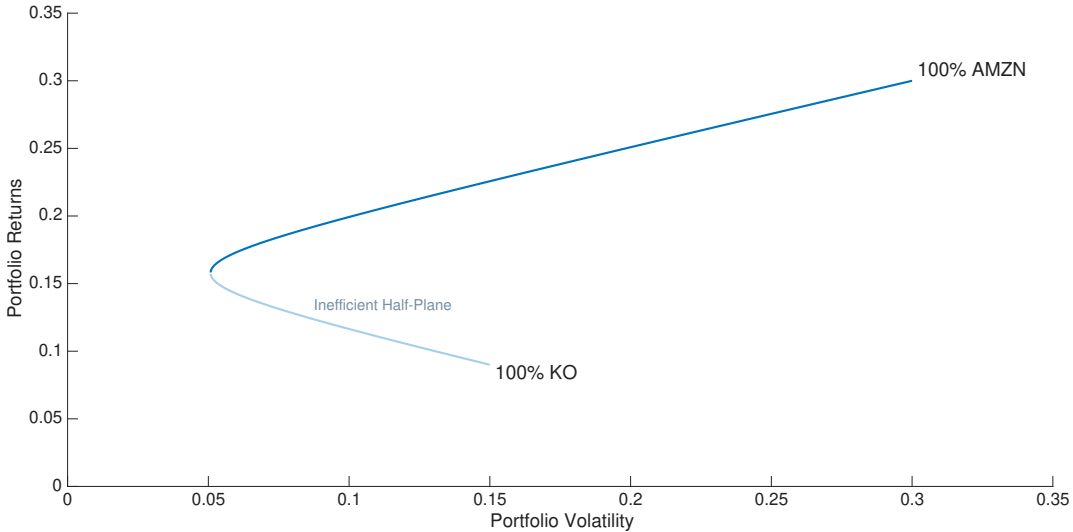
$$\sigma_p^2 = w^2\sigma_{KO}^2 + (1-w)^2\sigma_{AMZN}^2 + 2w(1-w)\rho\sigma_{KO}\sigma_{AMZN}$$
$$\sigma_p^2 = 0.354^2(0.15)^2 + 0.646^2(0.30)^2 + 2(0.354)(0.646)(-0.87)(0.15)(0.30) \approx 0.0225$$
$$\sigma_p = \sqrt{0.0225} = 15\%$$

The Efficient Frontier

KO had an average return of 9% and volatility of $\sigma_{\text{KO}} = 15\%$. A portfolio of 35.4% in KO and 64.6% in AMZN gives expected returns of 22.6% for the same level of volatility of 15%.

Hence, we say that investing in KO is dominated by a combination of KO and AMZN, and that it is inefficient to invest 100% in KO.

The Efficient Frontier



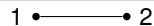
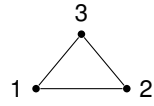
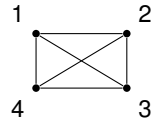
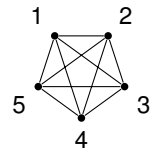
The Efficient Frontier



An inefficient portfolio is a portfolio in which it is possible to find another portfolio (using the same set of assets) that gives a higher expected return for the same level of risk.

A stock may be useful regardless of its own mean and variance due to its correlation to other stocks.

The Efficient Frontier with Many Assets

2 Assets		1 distinct **correlation**
3 Assets		3 distinct **correlations**
4 Assets		6 distinct **correlations**
5 Assets		10 distinct **correlations**

For N assets, there are $N(N-1)/2$ correlation pairs, so correlations entirely determine the portfolio variance when N is large.

The Efficient Frontier with Many Assets



See the file “Frontier Builder - Week 7.html”, along with “stock data - Week 7.csv” on K2.

Sharpe Ratio

The Sharpe Ratio of a portfolio is defined as

$$\text{Sharpe Ratio}_p = \frac{R_p - r^f}{\sigma_p}$$

It measures how much return you earn above the risk-free interest rate per unit of risk (volatility) in your portfolio.

The tangent portfolio is the portfolio with the highest Sharpe ratio in the efficient frontier of all assets.

The Market Portfolio



If we assume (among many other things) that all investors have identical beliefs about means, variances, and covariances across all assets, then everyone agrees about the composition of the tangent portfolio.

The tangent portfolio is the best investment in the investor's opportunity set. If everyone wants the same portfolio, by supply and demand, the composition of the market portfolio will exactly equal the composition of the tangent portfolio.

If every investor wants to hold 5% of their portfolio in Apple, the price of Apple stock will go up due to investor demand until it represents exactly 5% of the wealth in the economy.

What if I’m more risk-averse than the average investor?



If we only invest in stocks but we want less volatility in our portfolio, we will have to trade not just the mean, but also the Sharpe ratio of the portfolio (because the tangent portfolio has the highest).

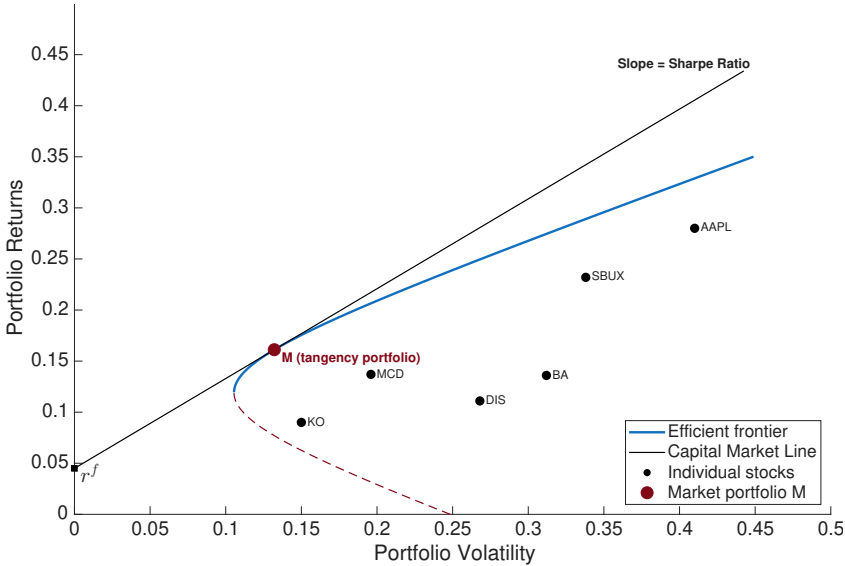
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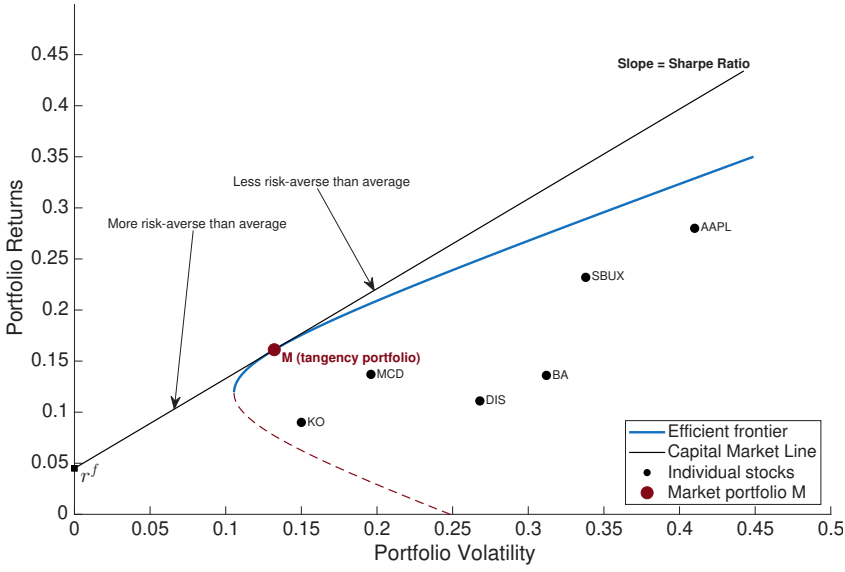
Instead of investing in a portfolio with a lower Sharpe ratio, we can invest in a portfolio of the risk-free asset (a government bond) and the market portfolio.

Two-Fund Separation Theorem. There exists two portfolios, the risk-free asset and the market portfolio, such that the optimal portfolio of any investor is a combination of these two assets.

The Capital Market Line



The Capital Market Line



Derivation of the CML (skip)

Consider a portfolio with two assets r^f and the market portfolio R^m. Assume the weight invested in the market portfolio is w. Then,

$$\mathbb{E}[R_p] = (1 - w)r^f + w\mathbb{E}[R^m]$$

Factoring, we get an equivalent representation,

$$\mathbb{E}[R_p] = r^f + w(\mathbb{E}[R^m] - r^f)$$

Find the variance of this portfolio. Assuming the variance on the risk-free asset is equal to zero (because it is assumed to be riskless), and the covariance between the risk-free asset and the market portfolio to be zero (it is not true in practice), we find that

$$\sigma_p = w\sigma^m$$

where σ^m is the volatility of the market portfolio. Using $w = \frac{\sigma_p}{\sigma^m}$,

$$\mathbb{E}[R_p] = r^f + \frac{\mathbb{E}[R^m] - r^f}{\sigma^m} \times \sigma_p$$

Derivation of the CML (skip)

The Capital Market Line is given by the equation

$$\mathbb{E}[R_p] = r^f + \frac{\mathbb{E}[R^m] - r^f}{\sigma^m} \times \sigma_p$$

where the slope is the Sharpe ratio on the market/tangent portfolio. To apply the CML, we either start with a target portfolio return or a target portfolio volatility, then identify the weights in each asset using

$$\mathbb{E}[R_p] = (1 - w)r^f + w\mathbb{E}[R^m]$$

based on the expected portfolio returns given the target portfolio vol or target portfolio returns.

Summary



What you need to know

- In an efficient portfolio, what matters is a stock's correlation/covariance to other stocks, not its individual mean and variance.
- Correlation is the covariance scaled to be a number between -1 and 1.
- The market portfolio is (theoretically) the portfolio with the highest Sharpe ratio in the opportunity set (highest excess return per unit of risk).
- All investors should hold a combination of the market portfolio and the risk-free asset, rather than individual stocks.

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Exercise 1 (slide 27, FADM ch. 6)



Suppose you buy 500 shares of Ford at $11 per share and 100 shares of Citigroup stock at $28 per share.

a. What is the weight invested in each asset?
   66.3% in Ford and 33.7% in Citi

Exercise 1 (slide 27, FADM ch. 6)



Suppose you buy 500 shares of Ford at $11 per share and 100 shares of Citigroup stock at $28 per share.

b. If Ford's share price goes up to $13 and Citigroup's rises to $40, what return did the portfolio earn?
   26.5%

Exercise 11.2 (slide 28, FADM ch. 6)



You own three stocks: 600 shares of Apple Computer, 10,000 shares of Cisco Systems, and 5000 shares of Colgate-Palmolive. The current share prices and expected returns of Apple, Cisco, and Colgate-Palmolive are, respectively, $547, $18, $95 and 12%, 10%, 8%.

a. What are the portfolio weights of the three stocks in your portfolio?
   33.4% in AAPL, 18.3% in CSCO and 48.31% in CL.

Exercise 11.2 (slide 28, FADM ch. 6)



You own three stocks: 600 shares of Apple Computer, 10,000 shares of Cisco Systems, and 5000 shares of Colgate-Palmolive. The current share prices and expected returns of Apple, Cisco, and Colgate-Palmolive are, respectively, $547, $18, $95 and 12%, 10%, 8%.

b. What is the expected return of your portfolio?
   9.7%

Exercise 11.2 (slide 28, FADM ch. 6)



You own three stocks: 600 shares of Apple Computer, 10,000 shares of Cisco Systems, and 5000 shares of Colgate-Palmolive. The current share prices and expected returns of Apple, Cisco, and Colgate-Palmolive are, respectively, \$547, \$18, \$95 and 12%, 10%, 8%.

c. Suppose the price of Apple stock goes up by \$20, Cisco rises by \$7, and Colgate-Palmolive falls by \$14. What are the new portfolio weights?
34.2% in AAPL, 25.1% in CSCO and 40.7% in CL.

Exercise 11.2 (slide 28, FADM ch. 6)



You own three stocks: 600 shares of Apple Computer, 10,000 shares of Cisco Systems, and 5000 shares of Colgate-Palmolive. The current share prices and expected returns of Apple, Cisco, and Colgate-Palmolive are, respectively, \$547, \$18, \$95 and 12%, 10%, 8%.

d. Assuming the stocks' expected returns remain the same, what is the expected return of the portfolio at the new prices?
   9.87%

Exercise 3 (slide 38, FADM ch. 6)

Security F has an expected value of 12 percent and a standard deviation of 9 percent per year. Security G has an expected return of 18 percent and a standard deviation of 25 percent per year.

a. What is the expected return on a portfolio composed of 30% of security F and 70% of security G?
   16.2%

Exercise 3 (slide 38, FADM ch. 6)

Security F has an expected value of 12 percent and a standard deviation of 9 percent per year. Security G has an expected return of 18 percent and a standard deviation of 25 percent per year.

b. If the correlation between the returns of security F and security G is 0.2, what is the standard deviation of this portfolio?
18.23%